excel project for an entry level data analyst for port folio

For an **entry-level Data Analyst**, your portfolio should demonstrate the skills recruiters expect:

- Excel formulas
- Data cleaning
- Pivot Tables
- Charts
- Dashboards
- Business insights

Here are the **5 best Excel portfolio projects**, ordered from easiest to most valuable.

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1. Sales Dashboard ⭐⭐⭐⭐⭐ (Best Project)

Skills Covered

- XLOOKUP
- INDEX + MATCH
- SUMIFS
- COUNTIFS
- Pivot Tables
- Pivot Charts
- Slicers
- Conditional Formatting
- Dashboard Design

Dataset

Contains:

- Order ID
- Order Date
- Customer
- Product
- Category

- Region
- Salesperson
- Quantity

ChatGPT Upgrade

- Sales
- Profit

Dashboard KPIs

- Total Sales
- Total Profit
- Total Orders
- Average Order Value

Charts

- Monthly Sales Trend
- Sales by Region
- Sales by Category
- Top 10 Products
- Top Salespeople

Business Questions

- Which region performs best?
- Which product generates the most revenue?
- Which month has the highest sales?
- Which salesperson performs best?

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2. HR Analytics Dashboard ⭐ ⭐ ⭐ ⭐☆

Dataset

- Employee ID
- Name
- Department
- Gender
- Salary
- Joining Date
- Age

- Experience
- Performance Rating
- Attrition

Dashboard

- Employee Count
- Average Salary
- Attrition Rate
- Department-wise Employees
- Gender Ratio
- Salary Distribution

Skills

- Pivot Tables
- Charts
- IF
- COUNTIFS
- SUMIFS
- Data Validation

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3. Retail Store Sales Analysis ⭐ ⭐ ⭐ ⭐☆

Dataset

- Product
- Store
- City
- Date
- Quantity
- Discount
- Sales
- Profit

Dashboard

- City-wise Sales
- Product-wise Profit
- Monthly Revenue

- Best Selling Products
- Top Stores

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4. Customer Analysis Dashboard ⭐⭐⭐⭐☆

Dataset

- Customer ID
- Customer Name
- Age
- Gender
- City
- Purchase Amount
- Category
- Purchase Date

Dashboard

- New Customers
- Returning Customers
- Top Customers
- Average Purchase
- Customer Segmentation

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5. Financial Budget Dashboard ⭐⭐⭐⭐☆

Dataset

- Department
- Budget
- Actual
- Variance
- Month

Dashboard

- Budget vs Actual
- Department Performance
- Monthly Spending

- Variance Analysis

Best Excel Functions to Include

Function	Purpose
XLOOKUP	Lookup values
INDEX + MATCH	Advanced lookup
SUMIFS	Multiple criteria totals
COUNTIFS	Count by criteria
IF	Logical conditions
IFERROR	Handle errors
TEXT	Format dates/text
LEFT, RIGHT, MID	Text extraction
UNIQUE	Remove duplicates
FILTER	Dynamic filtering
SORT	Sorting data
DATE functions	Time analysis

Excel Features Recruiters Like

- Excel Tables
- Named Ranges
- Data Validation
- Conditional Formatting
- Pivot Tables
- Pivot Charts

- Slicers
- Timelines
- Sparklines
- Dynamic Charts

Recommended Dashboard Layout

```
------------------------------------------------
                SALES DASHBOARD

KPI Cards
------------------------------------------
Total Sales | Profit | Orders | Avg Order

------------------------------------------
Monthly Sales Trend

------------------------------------------
Sales by Region

------------------------------------------
Sales by Category

------------------------------------------
Top 10 Products

------------------------------------------
Top Salespersons

Slicers:
✓ Year
✓ Region
✓ Category
✓ Salesperson
```

Project Workflow

1. Import the dataset.
2. Clean the data (remove duplicates, fix blanks, standardize formats).
3. Create calculated columns (Sales, Profit, etc.).
4. Build Pivot Tables.

5. Create Pivot Charts.
6. Add slicers and timelines.
7. Design an interactive dashboard.
8. Write key business insights.

How to Explain This Project in an Interview

You can structure your explanation like this:

1. **Project Overview:** "I built an interactive Sales Dashboard in Excel to analyze sales performance across regions, products, and salespeople."
2. **Data Preparation:** "I cleaned the data by removing duplicates, handling missing values, and formatting dates and numbers."
3. **Analysis:** "I used Pivot Tables, SUMIFS, XLOOKUP, and other Excel functions to calculate KPIs and summarize the data."
4. **Visualization:** "I created Pivot Charts, KPI cards, and slicers to make the dashboard interactive."
5. **Insights:** "The dashboard helps identify top-performing regions, products, and salespeople, as well as monthly sales trends."

My top recommendation

Since you've been learning Excel functions like **XLOOKUP, HLOOKUP, INDEX + MATCH, SUMIFS, and Pivot Tables**, I recommend building the **Sales Dashboard** first. It showcases almost every core Excel skill expected from an entry-level data analyst and is one of the strongest projects you can include in your portfolio.

After that, you can add:

1. Sales Dashboard
2. HR Analytics Dashboard
3. Retail Store Analysis

This combination gives you a well-rounded portfolio that demonstrates data cleaning, analysis, reporting, and dashboard creation.



If you're preparing for an **entry-level Data Analyst interview**, the **Sales Dashboard** is one of the strongest Excel portfolio projects. Here's a complete explanation you can use in your portfolio, resume, or interview.

Excel Sales Dashboard Project – Complete Explanation

Project Title

Sales Performance Dashboard in Microsoft Excel

Project Objective

The goal of this project is to analyze sales data and create an interactive dashboard that helps business users monitor sales performance, identify trends, and make better business decisions.

Business Problem

A retail company has thousands of sales transactions. Management wants to answer questions such as:

- What are the total sales and profits?
- Which region generates the highest revenue?
- Which products sell the most?
- Which salesperson performs the best?
- Which months have the highest sales?
- How are sales changing over time?

Instead of checking raw data manually, the company needs an interactive dashboard.

Dataset

The dataset contains approximately **5,000–10,000 sales records** with columns like:

Column	Description
Order ID	Unique order number

Column	Description
Order Date	Date of purchase
Customer Name	Customer who placed the order
Product	Product sold
Category	Product category
Region	Sales region
Salesperson	Employee who made the sale
Quantity	Units sold
Unit Price	Price per unit
Sales	Total sales amount
Profit	Profit earned

Data Cleaning

Before analysis, the data was cleaned by:

- Removing duplicate records
- Handling missing values
- Correcting date formats
- Formatting currency values
- Standardizing product and region names
- Checking for data entry errors

Excel Features Used

Formulas

- XLOOKUP
- INDEX + MATCH

- SUMIFS
- COUNTIFS
- IF
- IFERROR
- TEXT
- YEAR
- MONTH
- EOMONTH

Example

Calculate Total Sales

```excel
=Quantity*Unit Price
```

Region-wise Sales

```excel
=SUMIFS(Sales,Region,A2)
```

Pivot Tables

Created Pivot Tables to summarize:

- Sales by Region
- Sales by Category
- Sales by Product
- Monthly Sales
- Salesperson Performance

Pivot Charts

Used charts such as:

- Column Chart
- Line Chart
- Pie Chart

- Bar Chart

These charts make trends easy to understand.

Dashboard KPIs

The dashboard displays:

- Total Sales
- Total Profit
- Total Orders
- Average Order Value
- Top Product
- Best Salesperson

Interactive Features

Added:

- Slicers
  - Region
  - Category
  - Salesperson
- Timeline
  - Order Date

Users can filter the dashboard instantly without changing formulas.

Dashboard Layout

```
--------------------------------------------------
               SALES DASHBOARD

Total Sales      Total Profit      Total Orders

--------------------------------------------------

Monthly Sales Trend

--------------------------------------------------
```

```
Sales by Region

------------------------------------------------

Sales by Category

------------------------------------------------

Top 10 Products

------------------------------------------------

Top Salespersons

------------------------------------------------

Slicers
Region
Category
Salesperson
Year
```

Business Insights

The dashboard helps answer questions such as:

- Which region generates the highest revenue?
- Which products contribute most to sales?
- Which salesperson has the highest sales?
- Which months have peak sales?
- Which category is the most profitable?
- Are sales increasing or decreasing over time?

These insights help management make informed decisions about inventory, marketing, and sales strategy.

Skills Demonstrated

This project showcases:

- Data Cleaning
- Data Analysis

- Excel Formulas
- Lookup Functions
- Pivot Tables
- Pivot Charts
- Dashboard Design
- Data Visualization
- Business Analysis
- Reporting

Challenges Faced

Some common challenges included:

- Cleaning inconsistent data
- Handling missing values
- Creating dynamic Pivot Tables
- Designing a clear dashboard layout
- Making filters work smoothly with charts

These were addressed using Excel's data cleaning tools, structured tables, and slicers.

Outcome

The final dashboard provides an interactive view of sales performance, allowing users to filter data by region, category, salesperson, and date. It makes it easier to identify trends, compare performance, and monitor key business metrics in one place.

How to Explain This Project in an Interview (2–3 Minutes)

> "I developed a Sales Performance Dashboard in Microsoft Excel using a retail sales dataset. First, I cleaned the data by removing duplicates, handling missing values, and standardizing formats. I then used formulas such as XLOOKUP, SUMIFS, COUNTIFS, and IF to perform calculations and create helper columns. Next, I built Pivot Tables to summarize sales by region, category, product, and salesperson. I created Pivot Charts and designed an interactive dashboard with KPI cards, slicers, and a timeline. The dashboard allows users to filter data and quickly understand sales trends, top-performing products, profitable regions, and salesperson performance. This project

demonstrates my skills in Excel, data analysis, dashboard creation, and presenting business insights."

This explanation is suitable for **entry-level Data Analyst interviews** and clearly demonstrates both your technical Excel skills and your ability to derive meaningful business insights.



interview questions on sales dash board project

Here are the most common interview questions you may be asked about an **Excel Sales Dashboard Project**, along with sample answers.

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1. Tell me about your Sales Dashboard project.

Sample Answer

"I created an interactive Sales Dashboard in Microsoft Excel to analyze sales performance using a retail sales dataset. I cleaned the data, created calculated columns, built Pivot Tables and Pivot Charts, and designed an interactive dashboard with slicers. The dashboard helps users monitor KPIs such as Total Sales, Total Profit, Total Orders, and Average Order Value, and analyze performance by region, category, product, and salesperson."

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2. What was the objective of your project?

Sample Answer

"The objective was to convert raw sales data into an interactive dashboard that enables management to track business performance, identify sales trends, and make data-driven decisions."

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3. Where did you get the dataset?

Sample Answer

> "I used a publicly available retail sales dataset suitable for portfolio projects. I selected it because it included realistic business information such as products, customers, regions, quantities, sales, and profits."

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4. What data cleaning steps did you perform?

Sample Answer

- Removed duplicate records
- Checked for missing values
- Corrected date formats
- Standardized product and region names
- Converted numeric columns to the correct format
- Removed unnecessary spaces

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5. Which Excel functions did you use?

Sample Answer

I used:

- XLOOKUP
- INDEX + MATCH
- SUMIFS
- COUNTIFS
- IF
- IFERROR
- TEXT
- YEAR
- MONTH
- EOMONTH

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6. Why did you use Pivot Tables?

Sample Answer

> "Pivot Tables helped me quickly summarize thousands of sales records by region, category, product, and salesperson without writing complex formulas."

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7. What KPIs did you include?

Sample Answer

- Total Sales
- Total Profit
- Total Orders
- Average Order Value
- Top Product
- Best Salesperson

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8. How did you calculate Total Sales?

Sample Answer

If the dataset didn't already have Sales:

```excel
=Quantity*Unit Price
```

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9. How did you calculate Total Profit?

Sample Answer

```excel
=Sales-Cost
```

or

excel

```
=(Unit Price-Cost Price)*Quantity
```

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10. What charts did you use?

Sample Answer

- Column Chart
- Bar Chart
- Line Chart
- Pie Chart

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11. Why did you use slicers?

Sample Answer

> "Slicers make the dashboard interactive. Users can instantly filter all Pivot Tables and Pivot Charts by Region, Category, Salesperson, or Year."

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12. What business questions does your dashboard answer?

Sample Answer

- Which region has the highest sales?
- Which product sells the most?
- Which category is most profitable?
- Who is the top salesperson?
- Which month has the highest revenue?
- How are sales changing over time?

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13. What challenges did you face?

Sample Answer

- Duplicate data
- Missing values
- Incorrect date formats
- Connecting one slicer to multiple Pivot Tables
- Designing a clean dashboard layout

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14. How did you solve those challenges?

Sample Answer

> "I cleaned the data before analysis, converted the data into an Excel Table, refreshed Pivot Tables after updates, and connected slicers to all relevant Pivot Tables using Report Connections."

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15. Why did you choose Excel instead of Power BI?

Sample Answer

> "For this project, Excel was sufficient because it provides strong data analysis capabilities with Pivot Tables, formulas, charts, and dashboards. It's also widely used in many organizations. For larger datasets and advanced visualizations, I would consider Power BI."

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16. How many records did your dataset have?

Sample Answer

> "The dataset contained approximately 5,000–10,000 sales records, which was enough to demonstrate cleaning, analysis, and dashboard creation."

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17. Which Excel features did you use?

Sample Answer

- Excel Tables

- Pivot Tables
- Pivot Charts
- Slicers
- Conditional Formatting
- Named Ranges
- Data Validation
- Lookup Functions

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18. What insights did you find?

Sample Answer

- The South region generated the highest sales.
- Electronics was the highest-selling category.
- Sales peaked during the holiday season.
- A small number of products contributed a large share of revenue.
- Some regions had high sales but lower profits, indicating higher costs or discounts.

(Adjust these insights to match your actual dataset.)

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19. If your manager asks for Sales by Quarter, what would you do?

Sample Answer

> "I would group the Order Date field by Quarters in a Pivot Table or create a Quarter helper column using Excel formulas, then update the dashboard."

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20. How would you improve the dashboard?

Sample Answer

I would add:

- Profit Margin KPI
- Year-over-Year Growth
- Dynamic Top 10 Products

- Forecasting Charts
- Customer Segmentation
- Return Rate Analysis

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21. If new sales data is added, how will your dashboard update?

Sample Answer

> "I stored the source data as an Excel Table, so new rows are included automatically. Then I refresh the Pivot Tables and Pivot Charts to update the dashboard."

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22. What did you learn from this project?

Sample Answer

> "This project improved my skills in data cleaning, Excel formulas, Pivot Tables, dashboard design, and converting raw business data into meaningful insights. It also taught me how to present information in a way that supports business decisions."

Technical Follow-up Questions

An interviewer may also ask:

- What is the difference between **SUMIF** and **SUMIFS**?
- When would you use **XLOOKUP** instead of **VLOOKUP**?
- Why use **INDEX + MATCH**?
- What is a **Pivot Cache**?
- How do you refresh a Pivot Table?
- How do you connect one slicer to multiple Pivot Tables?
- What are calculated fields in Pivot Tables?
- How do you remove duplicates in Excel?
- What is Conditional Formatting?
- What are Excel Tables and why are they useful?

Tip for Entry-Level Interviews

Don't just explain **how** you built the dashboard. Also explain **why** it helps the business. Employers value analysts who can turn data into actionable insights, not just create charts. If you can explain the business impact of each KPI and visualization, you'll make a stronger impression.

why did you choose this project

This is a very common interview question. Your answer should connect the project to business value and your learning goals.

Interview Question:

Why did you choose this Sales Dashboard project?

Sample Answer (Best for Entry-Level Data Analyst)

> "I chose the Sales Dashboard project because it covers the core skills expected from an entry-level data analyst. It allowed me to work with a realistic sales dataset, clean and organize the data, perform analysis using Excel functions, create Pivot Tables and Pivot Charts, and build an interactive dashboard. Sales data is used by almost every business, so I wanted to solve a practical business problem and demonstrate how data can support decision-making."

Alternative Answer (More Professional)

> "I selected this project because sales analysis is one of the most common business use cases in data analytics. Through this project, I was able to practice the complete data analysis process—from data cleaning and transformation to visualization and reporting. It also helped me improve my skills in Excel formulas, Pivot Tables, and dashboard design while learning how to present insights in a way that supports business decisions."

If the interviewer asks, "Why not another project?"

You can answer:

> "I chose a Sales Dashboard because it combines multiple analytical skills in one project. Unlike a simple report, it requires data cleaning, KPI creation, trend analysis, interactive filtering, and visualization. It demonstrates both my technical Excel skills and my ability to communicate business insights."

Follow-up Question: What skills did this project help you develop?

You can say:

- Data cleaning and preparation
- Excel formulas (XLOOKUP, SUMIFS, IF, COUNTIFS)
- Pivot Tables and Pivot Charts
- Interactive dashboards using slicers
- KPI creation
- Trend analysis
- Data visualization
- Business insight generation
- Presenting findings to stakeholders

Strong 1-Minute Interview Answer

> "I chose the Sales Dashboard project because it reflects a real business scenario and demonstrates the complete workflow of a data analyst. I cleaned the sales data, analyzed it using Excel formulas and Pivot Tables, and built an interactive dashboard with KPIs and charts. This project strengthened my technical Excel skills while teaching me how to convert raw data into meaningful insights that can help management monitor performance and make better business decisions. I believe it is an excellent project for demonstrating the practical skills required for an entry-level Data Analyst role."

